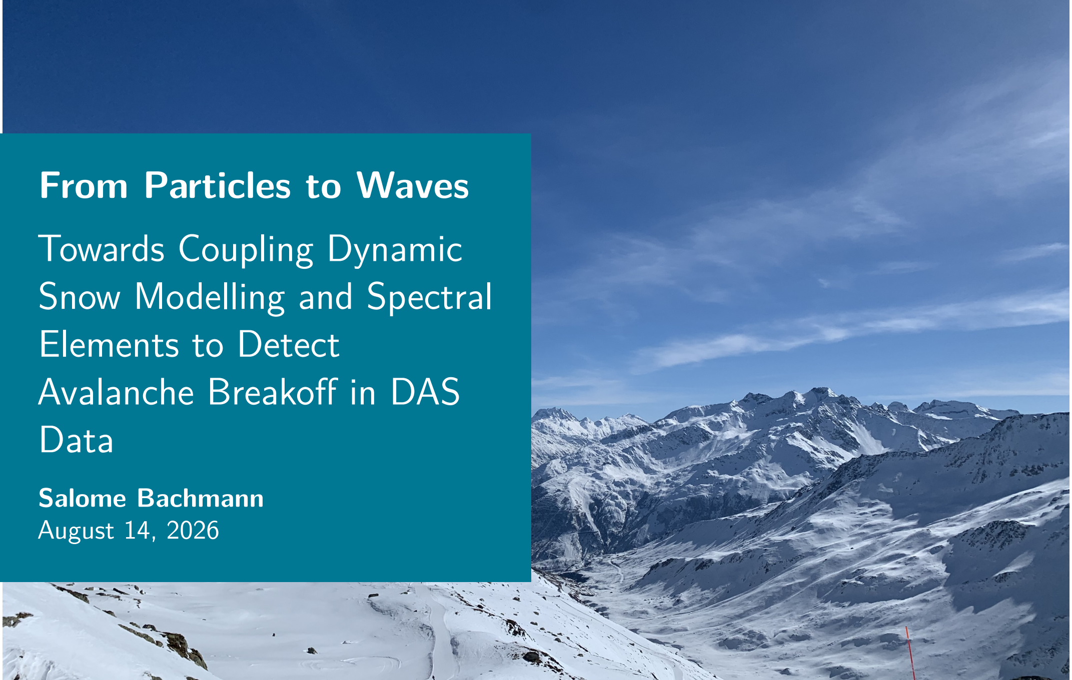


From Particles to Waves

Towards Coupling Dynamic Snow Modelling and Spectral Elements to Detect Avalanche Breakoff in DAS Data

Salome Bachmann

August 14, 2026

Why model crack propagation in snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weak layer (Gaume et al., 2017)

Relevance

- Crack propagation at different speeds: sub-Rayleigh (below vs of material) and Supershear (above vs) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation $\rightarrow$ critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

Overview

- **Goal**: Couple dynamic snow modelling with SE to model crack propagation in snow
- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations

Nomenclature & Abbreviations

- **SE, FE, FEM**: Spectral elements, Finite Element (Method), done in SALVUS
- **MPM**: Material Point Method Numerical Modeling
- **PST**: Propagation Saw Test
- **F-k**: Frequency-Wavenumber Plot
- **DAS**: Distributed Acoustic Sensing (Fibre-Optics)
- **STF**: Source-time function (the wavelet fired by SALVUS)

Research Questions

- What FE modelling framework is required to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a PST are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Simple Model I

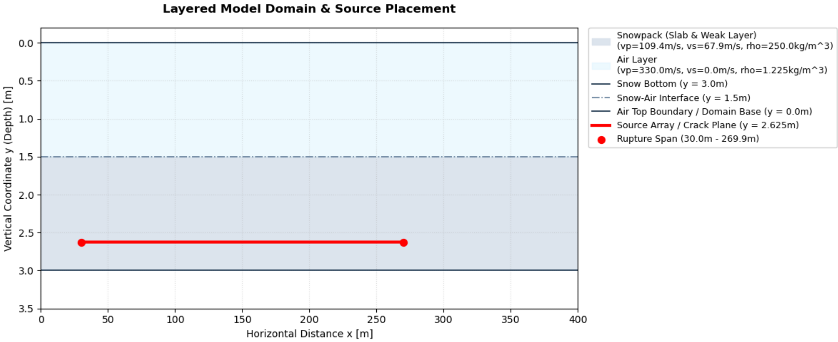


Figure: Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

Simple Model II

- Set up to test custom STF in SALVUS, see physical effects on output
- Sources spaced over 240m in 10cm intervals
- 20Hz central frequency Ricker wavelet
- Source Wavelet in xx (1), xy (1) and yy (-1) direction
- Time delay of wavelet calculated based on propagation speed: sub-Rayleigh speed is $0.6v_s$, Supershear $1.6v_s$ (Trottet et al., 2022)

Results

- SALVUS Output: Displacement field u
- $\frac{\partial u_i}{\partial t} \rightarrow$ velocity in x and y
- $\frac{\partial u_i}{\partial j} \rightarrow$ strain in x and y ϵ
- $\frac{\partial^2 u_i}{\partial t \partial t} \rightarrow$ strain rate $\dot{\epsilon}$
- Simulation of DAS data from $\epsilon, \ \dot{\epsilon}$ using Pyber Gauge length class by Noe, 2025, gauge length acoording to Turquet et al., 2024
- Spate-time plots of strain, strain rate and velocity at different reciever depths
- Frequency-wavenumber plots with theoretical speeds and dispersion curves
- Animations of velocity wavefields

Simple Sub-Rayleigh: Strain and Strain Rate I

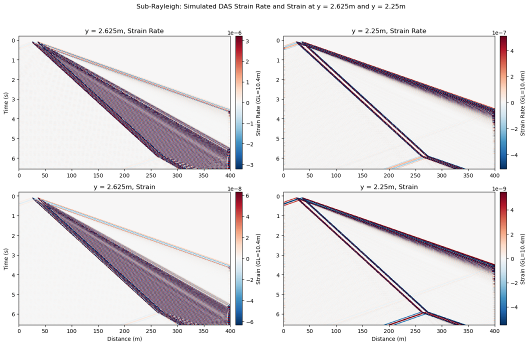


Figure: Horizontal Strain and strain rate for sub-Rayleigh crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

Simple Sub-Rayleigh: Strain and Strain Rate II

- Two-wavefront structure
- Main (thicker) wavefront: slope corresponds to prescribed rupture speed
- Second wavefront: coming only from first source
- Wavefront is emitted by all sources but they interfer (Madariaga, 1976), as shown by wider source spacing
- Slope of line close to p-wave speed in medium: S0 wave bounded by plate wave speed limit ("Waves in Plates", 2014)
- S0/P-wave precursor observed in field data by Herwijnen and Schweizer, 2011
- Oscillatory behavior of both wavefronts (polarity flip at adjacent timesteps)
- At rupture arrest (270m): v-shaped arrest signature: bidirectional, at fastest speed in medium (Freund & Freud, 1998)
- Significant weakening of amplitude between two depths

Simple Supershear: Strain and Strain Rate I

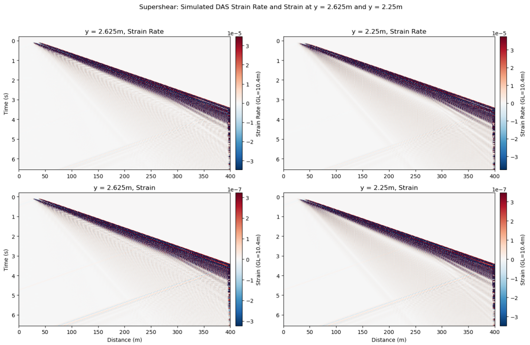


Figure: Horizontal Strain and strain rate for Supershear crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

Simple Supershear: Strain and Strain Rate II

- One wavefront, contrary to sub-Rayleigh
- Slope correspotnts to rupture speed (within 1m/s of vp and S0)
- Dispersion and broadening of main wavefront (Aki & Richards, 1981)
- Oscillatory wavefront
- At x=270m, unidirectional fan-like arrest front: rupture outruns vs, so arrest energy is only propagated forward (Madariaga, 1977)
- S0 precursor can be seen in traces
- Supershear larger amplitude compared to sub-Rayleigh

Simple model: Lamb Wave Identification I

- Polarity oscillation in all plots: indicative of Lamb waves
- Plotting dispersion curves over f-k spectra: energy is concentrated at symmetric (S0) and anitsymetric (A0) mode
- Antisymmetric mode gets excited preferentially when source is applied normal to midplane of material (Achenbach, 1973; Lamb, 1917; "Waves in Plates", 2014), so in this case yy
- Confirmed by animations of wavefield:

Simple model: Lamb Wave Identification II

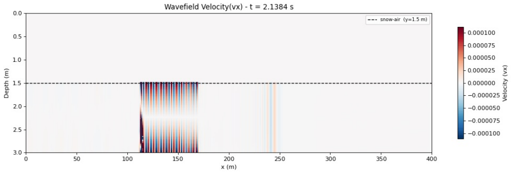


Figure: Animation still for Sub-Rayleigh Vx

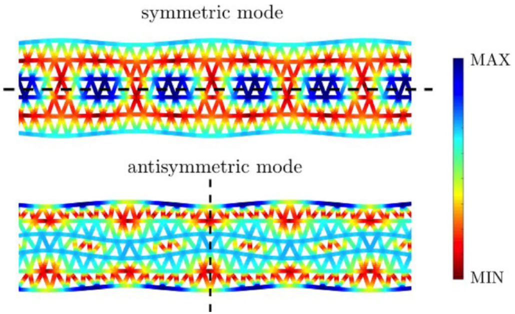


Figure: Lamb wave modes intensity, taken from Carta et al., 2023

MPM-Guided Model I

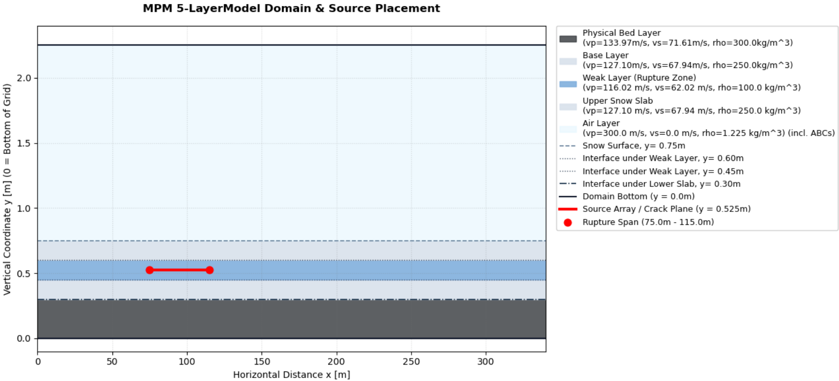


Figure: Five-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

MPM-Guided Model II

- Source-time functions obtained from MPM-simulation of PST by Trottet et al., 2022
- Source Wavelet in all three directions, but irregular and different for every source point

MPM-Guided: Strain and Strain Rate I

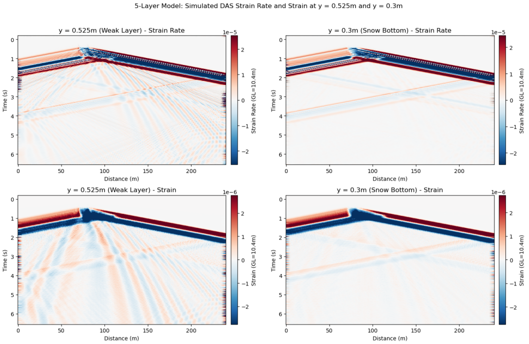


Figure: Horizontal Strain and strain rate for MPM-guided model. Plotted at crack depth (0.525) and at the snow bottom (0.3m). Gauge length is 10.4m.

MPM-Guided: Strain and Strain Rate II

- One two main wavefronts observed
- Crack extent from 75m to 125m: sub-Rayleigh speed from front
- Positive wedge in crack area: extends until rupture arrest, then jumps to same thickness as negative front $\rightarrow$ quasi-static offset
- Backward-propagating front and forward after crack arrest: close to vp/S0 from slope
- Second, weaker wavefront: at onset but also arrest: same arrest structure as sub-Rayleigh
- Oscillatory wavefronts

Quasi-Static Offset

- Static field represents permanent deformation after sources have fired (Archambeau, 1968; Zhu & Rivera, 2002)
- MPM STFS are not symmetric: contrary to Ricker wavelets, they do not integrate to 0 over time $\rightarrow$ net displacement of material (Aki & Richards, 1981; Zhu & Rivera, 2002)
- Cumulative integral of MPM STFs is negative: collapse of the weak layer, as observed by (Gaume et al., 2018; Heierli et al., 2008; Heierli et al., 2003; Trottet et al., 2022)
- All components of MPM STF (directionality) indicate mixed-mode anticrack instead of pure opening or shearing (mode I, II or II)

MPM-Guided Model: Lamb Wave Identification I

- Like in simple model: oscillatory wavefronts $\rightarrow$ A0 mode

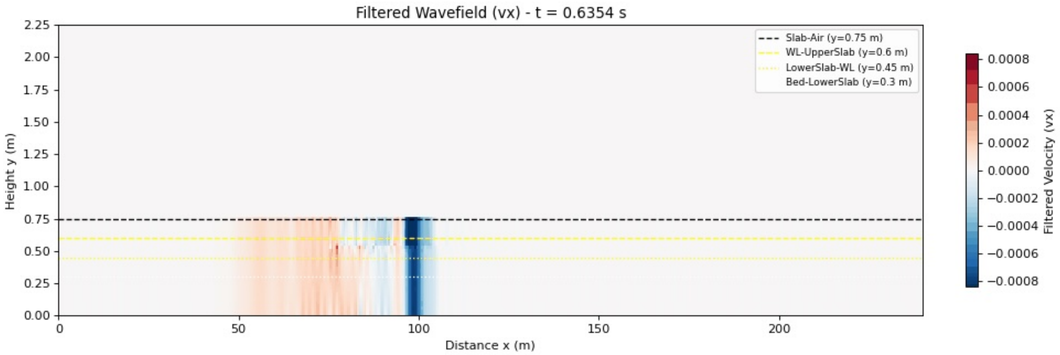


Figure: Animation still for MPM Vx

MPM-Guided Model: Lamb Wave Identification II

- F-k plots also show concentration of energy around A0 and S0 dispersion curves
- Polarity reversal around slab midplane in animation indicative of A0

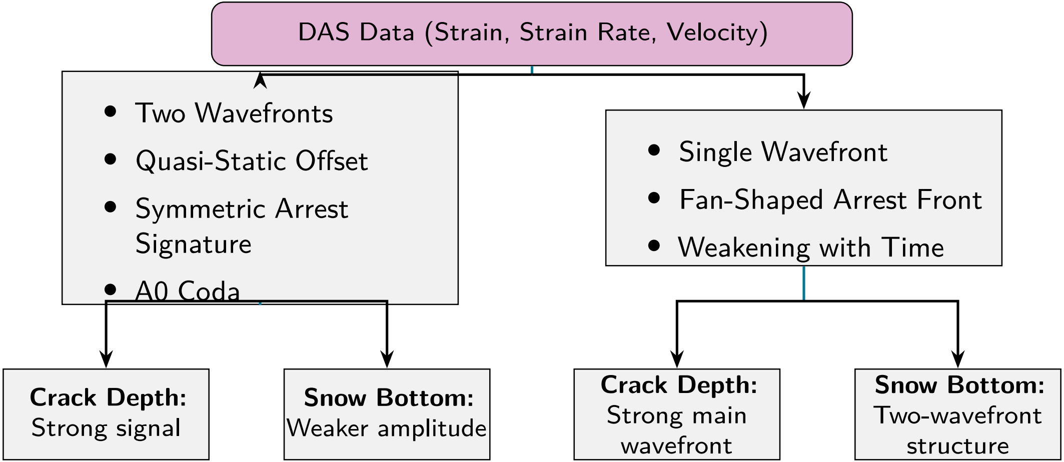
Summary of Results & Discussion I

- S0 Precursor observed in all Strain & Strain Rate Plots, as well as in traces
- Main wavfront in all cases is A0 $\rightarrow$ source normal to slab midplane
- Different rupture arrest structures depending on propagation speed
- Quasi-static wavefield for sources that do not integrate to 0 over time
- Weakening of amplitude only in simple case $\rightarrow$ also thicker slab

Implications for DAS field data I

- Identification of rupture speed via number of wavefronts & arrest structure
- Precursors in sub-Rayleigh: S0 and Quasi-Static offset
- Consistent with observations by Herwijnen and Schweizer, 2011
- Structures identified as being geometry-independent
- Signals identified as being larger than instrument noise of Silixa iDAS used by Paitz et al., 2023
- Supershear signals larger in amplitude: possibly false conclusions if transition happens
- Identification flowchart:

Implications for DAS field data II



Implications for DAS field data III

Figure: Graphical Representation to Identify Different Signatures in DAS

Outlook

- Add topography to models
- Verify Supershear effects for more complex STFs as well
- Inhomogeneous, multi-layered snowpack
- Transitional propagation regimes
- Comparison to field data $\rightarrow$amplitude of signals compared to noise
- 3D model

Answer to Research Questions - Conclusions

- **What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?** Rupture-arrest signatures depending on propagation speed, S0 precursor, A0 main wavefront for both propagation regimes. Changes in amplitude
- **To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?** Verification of sub-Rayleigh S0 precursor, A0 mode excitation, changes because appearance of quasi-static offset
- **How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?** Precursors (S0 and quasi-static offset) used as "early" warning, identification of propagation speed using arrest structures, possibility of identifying transitions by number of wavefronts

Questions?

Bibliography I

Achenbach, J. (1973). *Wave Propagation in Elastic Solids*. Elsevier.

Aki, K., & Richards, P. G. (1981).K. Aki & P. G. Richards 1980. Quantitative Seismology, Theory and Methods. Volume I: 557 pp., 169 illustrations. Volume II: 373 pp., 116 illustrations. San Francisco: Freeman. Price: Volume I, U.S. 35.00. ISBN 0 7167 1058 7 (Vol. I), 0 7167 1059 5 (Vol. II).. *Geological Magazine*, *118*(2), 208–208. https://doi.org/10.1017/S0016756800034439

Archambeau, C. B. (1968).General theory of elastodynamic source fields. *Reviews of Geophysics*, *6*(3), 241–288. https://doi.org/10.1029/RG006i003p00241

Bergfeld, B., Gaume, J., Bobillier, G., Pellet, A., Schweizer, J., & van Herwijnen, A. (2025).Signatures of the sub-Rayleigh to supershear fracture transition in snow avalanche experiments. *Nature Communications*, *16*(1), 11153. https://doi.org/10.1038/s41467-025-65825-6

Bibliography II

Carta, G., Nieves, M. J., & Brun, M. (2023).Lamb waves in discrete homogeneous and heterogeneous systems: Dispersion properties, asymptotics and non-symmetric wave propagation. *European Journal of Mechanics - A/Solids*, *100*, 104695. https://doi.org/10.1016/j.euromechsol.2022.104695

Edme, P., Paitz, P., Walter, F., van Herwijnen, A., & Fichtner, A. (2023, February). Fiber-optic detection of snow avalanches using telecommunication infrastructure. https://doi.org/10.48550/arXiv.2302.12649

Freund, L. B., & Freud, L. B. (1998, March). *Dynamic Fracture Mechanics*. Cambridge University Press.

Gaume, J., Gast, T., Teran, J., van Herwijnen, A., & Jiang, C. (2018).Dynamic anticrack propagation in snow. *Nature Communications*, *9*(1), 3047. https://doi.org/10.1038/s41467-018-05181-w

Bibliography III

Gaume, J., van Herwijnen, A., Chambon, G., Birkeland, K. W., & Schweizer, J. (2015).Modeling of crack propagation in weak snowpack layers using the discrete element method. *The Cryosphere*, *9*(5), 1915–1932. https://doi.org/10.5194/tc-9-1915-2015

Gaume, J., van Herwijnen, A., Chambon, G., Wever, N., & Schweizer, J. (2017).Snow fracture in relation to slab avalanche release: Critical state for the onset of crack propagation. *The Cryosphere*, *11*(1), 217–228. https://doi.org/10.5194/tc-11-217-2017

Heierli, J., Gumbsch, P., & Zaiser, M. (2008).Anticrack Nucleation as Triggering Mechanism for Snow Slab Avalanches. *Science*, *321*(5886), 240–243. https://doi.org/10.1126/science.1153948

Heierli, J., van Herwijnen, A., Gumbsch, P., & Zaiser, M. (2003).Anticracks: A new theory of fracture initiation and fracture propagation in snow.

Herwijnen, A. V., & Schweizer, J. (2011). Seismic sensor array for monitoring an avalanche start zone: Design, deployment and preliminary results. *Journal of Glaciology*, *57*(202), 267–276. https://doi.org/10.3189/002214311796405933

Lamb, H. (1917). On waves in an elastic plate. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, *93*(648), 114–128. https://doi.org/10.1098/rspa.1917.0008

Madariaga, R. (1976). Dynamics of an expanding circular fault. *Bulletin of the Seismological Society of America*, *66*(3), 639–666. https://doi.org/10.1785/BSSA0660030639

Madariaga, R. (1977). High-frequency radiation from crack (stress drop) models of earthquake faulting. *Geophysical Journal International*, *51*(3), 625–651. https://doi.org/10.1111/j.1365-246X.1977.tb04211.x

Noe, S. (2025). Frontiers of full-waveform seismology: From global tomography to distributed and integrated fibre-optic strain sensing. https://doi.org/10.3929/ETHZ-C-000788455

Bibliography V

Paitz, P., Lindner, N., Edme, P., Huguenin, P., Hohl, M., Sovilla, B., Walter, F., & Fichtner, A. (2023). Phenomenology of Avalanche Recordings From Distributed Acoustic Sensing. *Journal of Geophysical Research: Earth Surface*, *128*(5), e2022JF007011. https://doi.org/10.1029/2022JF007011

Schweizer, J., Bruce Jamieson, J., & Schneebeli, M. (2003). Snow avalanche formation. *Reviews of Geophysics*, *41*(4). https://doi.org/10.1029/2002RG000123

SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.

Trottet, B., Simenhois, R., Bobillier, G., Bergfeld, B., van Herwijnen, A., Jiang, C., & Gaume, J. (2022). Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches. *Nature Physics*, *18*(9), 1094–1098. https://doi.org/10.1038/s41567-022-01662-4

Turquet, A., Wuestefeld, A., Svendsen, G. K., Nyhammer, F. K., Nilsen, E. L., Persson, A. P.-O., & Refsum, V. (2024). Automated Snow Avalanche Monitoring and Alert System Using Distributed Acoustic Sensing in Norway. *GeoHazards*, *5*(4), 1326–1345. https://doi.org/10.3390/geohazards5040063

Bibliography VI

Waves in Plates. (2014). In J. L. Rose (Ed.), *Ultrasonic Guided Waves in Solid Media* (pp. 76–106). Cambridge University Press. https://doi.org/10.1017/CBO9781107273610.008

Zhu, L., & Rivera, L. A. (2002).A note on the dynamic and static displacements from a point source in multilayered media. *Geophysical Journal International*, *148*(3), 619–627. https://doi.org/10.1046/j.1365-246X.2002.01610.x

MPM STF Workflow

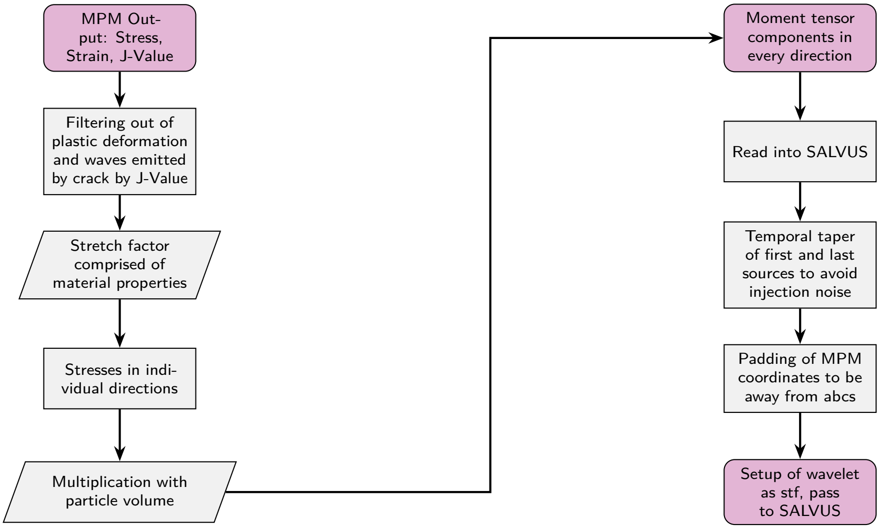


Figure: Workflow for conversion of MPM data and input as SALVUS STF.

Simple model velocity plots I

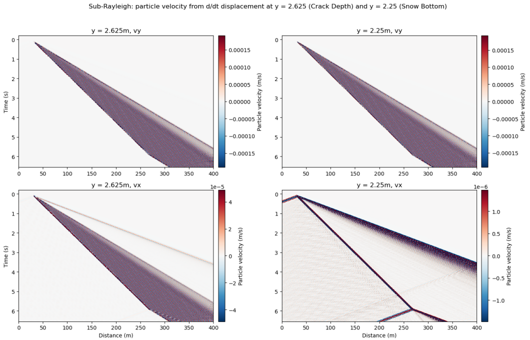


Figure: Simple model Sub-Rayleigh velocity plots. Crack depth on left side, snow bottom on right. Vy at top, vx at bottom. Velocity derived from displacement.

Simple model velocity plots II

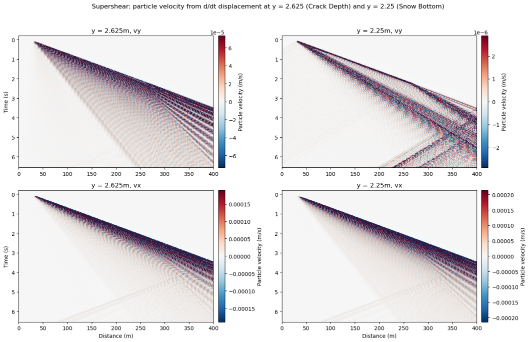


Figure: Simple model Supershear velocity plots. Crack depth on left side, snow bottom on right. Vy at top, vx at bottom. Velocity derived from displacement.

MPM velocity plots

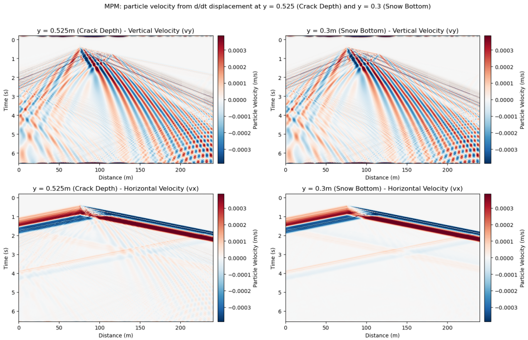
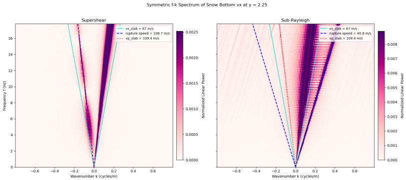


Figure: MPM-guided model velocity plots. Crack depth on left side, snow bottom on right. Vy at top, vx at bottom. Velocity derived from displacement.

Simple model F-k plots and dispersion curves



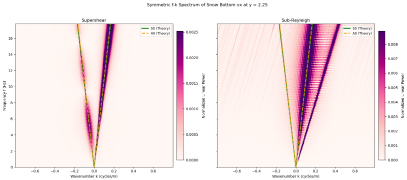
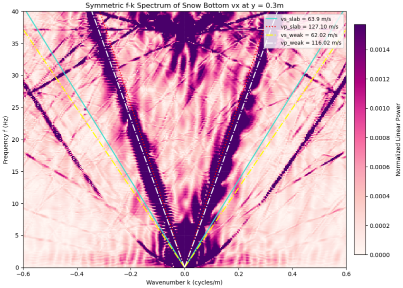


Figure: Supershear and Sub-Rayleigh F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM F-k plots



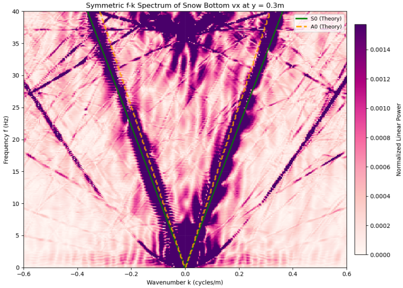


Figure: MPM F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM STF cumulative Integral

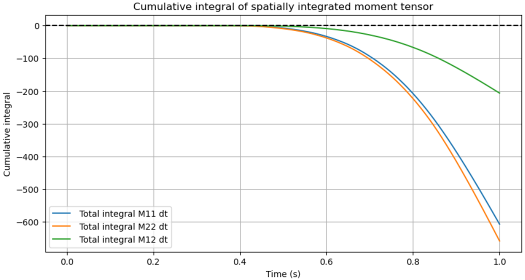


Figure: Cumulative integral over time of all MPM STF components

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